

■ Learning Check

1. The physical property possessed by certain particles that is responsible for electrical and magnetic phenomena is _____.
2. (True or False.) All three principal types of particles found in atoms are electrically charged.

3. A positive ion is formed when
 - (a) a neutral atom gains an electron.
 - (b) a neutral atom loses an electron.
 - (c) a negative ion loses an electron.
 - (d) All of the above.

ANSWERS: 1. electric charge 2. False 3. (b)

■ Learning Check

1. Like charges _____; unlike charges _____.
2. Two charged objects exert forces on each other. The sizes of both forces will increase if
 - (a) the objects are brought closer together.
 - (b) the charge on one of them is increased.
 - (c) the charges on both of them are increased.
 - (d) All of the above.

3. A charged particle creates an electric _____ in the space around it.
4. (True or False.) An electrostatic precipitator uses an electric field to make it rain.

ANSWERS: 1. repel, attract 2. (d) 3. field 4. False

■ Learning Check

1. An electric current can consist of
 - (a) positive ions flowing in a liquid.
 - (b) negative ions flowing in a liquid.
 - (c) electrons moving through a wire.
 - (d) All of the above.
2. A substance that is neither a good conductor nor a good insulator is called a _____.

3. What is the distinction between an excellent conductor, such as a copper wire used in an electronic device, and a superconductor?

ANSWERS: 1. (d) 2. semiconductor 3. small resistance versus zero resistance

■ Learning Check

1. (True or False.) A battery stores electric charge in the same way that a bottle stores water.
2. Which of the following is not true about voltage?
 - (a) It is analogous to fluid pressure.
 - (b) It equals the energy per unit charge.
 - (c) It equals the work that can be done per unit charge.
 - (d) Its unit of measure is equivalent to a joule per ampere.

3. The size of the current flowing through a conductor equals the _____ divided by the _____.
4. (True or False.) In a parallel circuit, the size of the current flowing through each device may be different.

ANSWERS: 1. False 2. (d) 3. voltage, resistance 4. True

■ Learning Check

1. An automobile headlight and a dashboard light use the same voltage, but the power input to the headlight is much larger because
 - (a) its resistance is lower.
 - (b) the current in it is larger.
 - (c) it uses energy faster.
 - (d) All of the above.
2. Name one or more devices in which ohmic heating is desirable.
3. (True or False.) A fuse is designed to stop the flow of charge in a circuit if the current is too large.
4. The standard unit of energy used in electric utility bills is the _____.

ANSWERS: 1. (d) 2. toaster, electric heater 3. True 4. kilowatt-hour

■ Learning Check

1. Direct current (DC)
 - (a) is produced by batteries.
 - (b) is used by most electric utilities.
 - (c) carries the audio signal to speakers.
 - (d) oscillates.
2. (True or False.) An incandescent lightbulb normally used with AC would not work if connected to DC (with the proper voltage).
3. AC voltages can be easily increased or decreased by using a _____.

ANSWERS: 1. (a) 2. False 3. transformer